

Project Title

Design of a Scalable and Secure Cloud Infrastructure on Google Cloud Platform (GCP)

Objective

Design a scalable, secure, and highly available cloud architecture capable of hosting a modern web application. The infrastructure should support increasing user traffic, ensure data security, and provide reliable performance.

Problem Statement

Traditional applications hosted on a single server often face challenges such as:

- Poor scalability during high traffic
- Single points of failure
- Limited availability and reliability
- Difficult maintenance and monitoring

Cloud computing addresses these challenges by providing on-demand resources, automatic scaling, and managed services.

Proposed Solution

Design a cloud architecture on Google Cloud Platform (GCP) that includes:

- Load balancing
- Auto-scaling application servers
- Managed database services
- Secure networking
- Storage and backup services
- Monitoring and logging

Architecture Components

1. User Layer

Users access the application through a web browser or mobile application.

2. Load Balancer

Distributes incoming requests across multiple application servers to ensure high availability and better performance.

3. Compute Layer

Virtual Machines (VMs) hosted in an Auto Scaling Group automatically increase or decrease based on traffic demand.

4. Database Layer

Managed Cloud SQL service stores application data securely and provides automated backups.

5. Storage Layer

Cloud Storage stores static files, images, documents, and backups.

6. Monitoring and Logging

Cloud Monitoring and Cloud Logging track system health, resource usage, and application performance.

Architecture Diagram

Users



Load Balancer



Auto Scaling Group (VM Instances)

Application Server 1

Application Server 2

Application Server



Cloud SQL Database



Cloud Storage (Backups & Static Files)

Monitoring & Logging Services

Key Features

- High Availability
- Automatic Scaling

- Secure Networking
- Managed Database Services
- Backup and Disaster Recovery
- Performance Monitoring
- Cost Optimization

Implementation Steps

Step 1: Create a VPC Network

Configure a Virtual Private Cloud (VPC) and subnets.

Step 2: Deploy Compute Instances

Create virtual machines and install the web application.

Step 3: Configure Load Balancer

Set up a load balancer to distribute traffic.

Step 4: Enable Auto Scaling

Create instance templates and configure scaling policies.

Step 5: Set Up Database

Deploy Cloud SQL and configure application connectivity.

Step 6: Configure Storage

Create Cloud Storage buckets for static files and backups.

Step 7: Enable Monitoring

Configure monitoring dashboards, logs, and alerts.

Advantages

- Handles large numbers of users efficiently
- Reduces downtime and service interruptions
- Improves application security
- Simplifies management and maintenance

- Supports future growth and expansion

Project Deliverables

Folder Structure

Cloud-Architecture-Design/

```
|  
|— README.md  
|— Project_Report.pdf  
|— Architecture_Diagram.png  
|— Implementation_Steps.docx  
|— Screenshots/  
|— References/
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Expected Outcome

A professional, portfolio-ready cloud architecture design demonstrating:

- Cloud infrastructure planning skills
- Knowledge of scalability and high availability
- Understanding of networking and security concepts
- Ability to document and present a complete cloud solution

Conclusion

This project presents a scalable and secure cloud architecture using Google Cloud Platform. The design follows cloud computing best practices by implementing load balancing, auto-scaling, managed databases, and monitoring services, making it suitable for real-world web applications and an excellent addition to a professional portfolio.